AI-Assisted Feature Identification for Continuous Reverse Engineering of UML Models

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Need for the Topic

Modern software projects follow Agile and CI/CD practices, causing source code to evolve rapidly. However, UML documentation often becomes outdated because developers rarely update diagrams manually. Frameworks like UMLRev address this by automatically generating UML diagrams from execution traces. Despite this advancement, UMLRev still requires manual feature identification before reverse engineering can begin. This manual process reduces automation, limits scalability, and increases developer effort. The proposed work aims to eliminate this bottleneck by integrating Artificial Intelligence to automatically identify software features, making continuous reverse engineering more efficient and practical for large-scale software projects.

Detailed Problem Statement

UMLRev successfully automates UML diagram generation but depends on developers to manually associate integration tests and execution traces with software features. This manual mapping is time-consuming, error-prone, and difficult to maintain as projects grow.

The proposed research focuses on improving this specific component by developing an AI-assisted feature identification module. Using techniques such as Large Language Models (LLMs) and code analysis, the system will automatically identify software features from commit messages, source code changes, method names, test cases, and project metadata. These identified features will then be linked to execution traces before UML generation.

The objective is to increase automation, reduce manual intervention, improve scalability, and generate more accurate feature-oriented UML documentation.

Methodology (Steps)

1. Developer commits source code changes.
2. CI/CD pipeline triggers the build and integration tests.
3. AI analyzes commit messages, changed files, method names, and test cases to identify software features.
4. AspectJ captures execution traces during integration testing.
5. AI maps execution traces to the identified software features.
6. UMLRev processes the feature-specific execution traces.
7. The Diagram Engine automatically generates updated UML Class and Sequence diagrams.
8. The generated documentation is integrated into the project repository.

9. Documentation remains continuously synchronized with software changes while requiring minimal manual effort.

Functionalities

- Automatic software feature identification using AI.
- Intelligent mapping of execution traces to software features.
- Continuous generation of UML Class and Sequence diagrams.
- Integration with CI/CD pipelines.
- Reduction of manual documentation effort.
- Improved scalability for large software projects.
- More accurate and context-aware UML documentation.
- Better maintainability and developer productivity.

References

1. UMLRev: A Context-Aware Feature-Oriented Approach to Support Continuous Reverse Engineering of UML Models
2. Object Management Group (OMG). *Unified Modeling Language (UML) Version 2.5.1 Specification*
3. StarUML Documentation. *StarUML User Guide and Documentation*
4. Martin Fowler – Continuous Integration
5. AspectJ Documentation
6. Eclipse UML2 Project Documentation